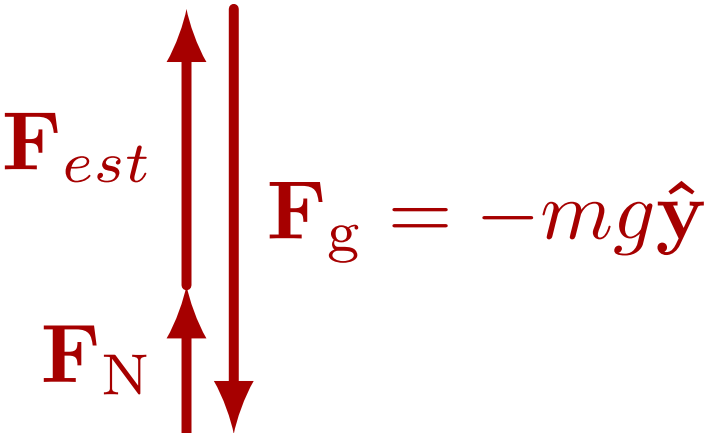
$\mathbf{F}_{est}$
$\mathbf{F}_{\mathrm{g}} = -mg\hat{\mathbf{y}}$
$\mathbf{F}_{\mathrm{N}}$